

Introduction to Convolutions with TensorFlow

experiment Lab schedule 50 minutes universal_currency_alt 5 Credits

show_chart Intermediate

★★★★★ [Rate Lab](#)

info This lab may incorporate AI tools to support your learning.

GSP632



Overview

A convolution is a filter that passes over an image, processes it, and extracts features that show a commonality in the image. In this lab you explore convolution filters. You learn what they are and how they work by processing an image to extract features from it! You also explore pooling, which compresses your image and further emphasizes the features.

Objectives

In this lab, you will learn how to:

- Load and draw an image from scipy, an open source Python library used for scientific and technical computing
- Create a filter as a 3x3 array and a convolution and see the effects on the image
- Run a pooling to see how it affects the output

Prerequisites

Although this is a self-standing lab, to maximize your learning consider taking these labs before taking this one:

- TensorFlow: Qwik Start
- Introduction to Computer Vision with TensorFlow

Setup and requirements

Before you click the Start Lab button

Read these instructions. Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources are made available to you.

This hands-on lab lets you do the lab activities in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials you use to sign in and access Google Cloud for the duration of the lab.

To complete this lab, you need:

- Access to a standard internet browser (Chrome browser recommended).

Note: Use an Incognito (recommended) or private browser window to run this lab. This prevents conflicts between your personal account and the student account, which may cause extra charges incurred to your personal account.

- Time to complete the lab—remember, once you start, you cannot pause a lab.

Note: Use only the student account for this lab. If you use a different Google Cloud account, you may incur charges to that account.

How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. If you need to pay for the lab, a dialog opens for you to select your payment method. On the left is the Lab Details pane with the following:

- The Open Google Cloud console button

- Time remaining
- The temporary credentials that you must use for this lab
- Other information, if needed, to step through this lab

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).

The lab spins up resources, and then opens another tab that shows the Sign in page.

Tip: Arrange the tabs in separate windows, side-by-side.

Note: If you see the **Choose an account** dialog, click **Use Another Account**.

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog.

"Username"

content_c

You can also find the Username in the Lab Details pane.

4. Click **Next**.

5. Copy the **Password** below and paste it into the **Welcome** dialog.

"Password"

content_c

You can also find the Password in the Lab Details pane.

6. Click **Next**.

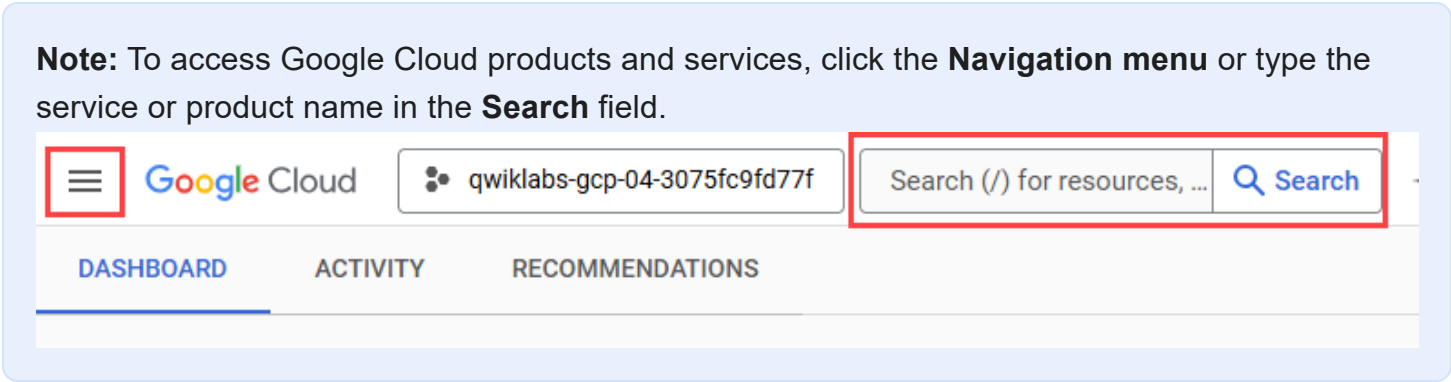
Important: You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.

Note: Using your own Google Cloud account for this lab may incur extra charges.

7. Click through the subsequent pages:

- Accept the terms and conditions.
- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.

After a few moments, the Google Cloud console opens in this tab.



Task 1. Open the notebook in Vertex AI Workbench

1. In the Google Cloud console, on the **Navigation menu** () , click **Vertex AI > Workbench**.
2. Find the **Workbench instance name** instance and click on the **Open JupyterLab** button.

The JupyterLab interface for your Workbench instance opens in a new browser tab.

Task 2. Navigate to the lab notebook

1. Click on the **notebook name** file.
2. In the **Select Kernel** dialog, choose **Python 3** from the list of available kernels.
3. Continue the lab in the notebook, and run each cell by clicking the **Run** icon at the top of the screen.
Alternatively, you can execute the code in a cell with **SHIFT + ENTER**.

Read the narrative and make sure you understand what's happening in each cell.

Click **Check my progress** to verify the objective.



Run the notebook

Check my progress

Congratulations!

This concluded the self-paced lab, Introduction to Convolutions with TensorFlow. You launched the convolutions notebook and explored convolutions and pooling.

Next steps / learn more

- Learn more about TensorFlow at [Getting started: training and prediction with TensorFlow Estimator](#).

Google Cloud training and certification

...helps you make the most of Google Cloud technologies. Our classes include technical skills and best practices to help you get up to speed quickly and continue your learning journey. We offer fundamental to advanced level training, with on-demand, live, and virtual options to suit your busy schedule. Certifications help you validate and prove your skill and expertise in Google Cloud technologies.

Manual Last Updated February 20, 2025

Lab Last Tested February 20, 2025

Copyright 2025 Google LLC All rights reserved. Google and the Google logo are trademarks of Google LLC. All other company and product names may be trademarks of the respective companies with which they are associated.